

For the use of a Registered Medical Practitioner only

PRESCRIBING INFORMATION

TALIS 20

(Tadalafil Tablets USP 20 mg)

COMPOSITION

Each TALIS 20 mg tablet contains 20 mg Tadalafil.

List of excipients

Poloxamer (188) (Lutrol F 68), Hydroxy Propyl cellulose, Lactose Monohydrate, Croscopolvidone, Microcrystalline cellulose, Croscopolvidone, Microcrystalline cellulose, Colloidal Anhydrous Silica, Magnesium Stearate

DESCRIPTION¹

TALIS 20 is a Yellow colored, oval shaped, biconvex, film coated tablets plain on both the sides.

TALIS 20 contains tadalafil which is a selective, reversible inhibitor of cyclic guanosine monophosphate (cGMP)-specific phosphodiesterase type 5 (PDE5).

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES¹

Pharmacodynamics

Mechanism of Action

Tadalafil is a selective, reversible inhibitor of cyclic guanosine monophosphate (cGMP)-specific phosphodiesterase type 5 (PDE5). When sexual stimulation causes the local release of nitric oxide, inhibition of PDE5 by tadalafil produces increased levels of cGMP in the corpus cavernosum. This results in smooth muscle relaxation and inflow of blood into the penile tissues, thereby producing an erection. Tadalafil has no effect in the treatment of erectile dysfunction (ED) in the absence of sexual stimulation.

Pharmacodynamic Effects

Tadalafil is a selective inhibitor of PDE5. PDE5 is an enzyme found in the smooth muscle of the corpus cavernosum, prostate and bladder as well as in vascular and visceral smooth muscle, skeletal muscle, platelets, kidney, lung, cerebellum and pancreas. The effect of tadalafil is more potent on PDE5 than on other phosphodiesterases.

Pharmacokinetics

Tadalafil is readily absorbed after oral administration and the mean maximum observed plasma concentration (C_{max}) is reported to be achieved at a median time of 2 hours after dosing. Absolute bioavailability of tadalafil following oral dosing has not been determined. The rate and extent of absorption of tadalafil are not influenced by food, thus tadalafil may be taken with or without food. The time of dosing (morning versus evening) had no clinically relevant effects on the rate and extent of absorption.

The mean volume of distribution is approximately 63L, indicating that tadalafil is distributed into tissues. At therapeutic concentrations, 94% of tadalafil in plasma is reported to be bound to proteins. Protein binding is not affected by impaired renal function. Less than 0.0005% of the administered dose was reported to appear in the semen of healthy individuals.

Tadalafil is predominantly metabolised by the cytochrome P450 (CYP) 3A4 isoform. The major circulating metabolite is the methylcatechol glucuronide. This metabolite is at least 13,000-fold less potent than tadalafil for PDE5. Consequently, it is not expected to be clinically active at reported metabolite concentrations.

Reportedly, the mean oral clearance for tadalafil is 2.5L/h and the mean half-life is 17.5 hours in healthy individuals. Tadalafil is reported to be excreted predominantly as inactive metabolites, mainly in the faeces (approximately 61% of the dose) and to a lesser extent in the urine (approximately 36% of the dose).

Tadalafil pharmacokinetics are reported to be linear with respect to time and dose. Over a dose range of 2.5mg to 20mg, exposure (AUC) increases proportionally with dose. Steady-state plasma concentrations are attained within 5 days of once-daily dosing. Pharmacokinetics determined with a population approach in patients with ED are reported to be similar to pharmacokinetics in subjects without ED.

Elderly

It was reported that, healthy elderly individuals (65 years or over), had a lower oral clearance of tadalafil, resulting in 25% higher exposure (AUC) relative to healthy individuals aged 19 to 45 years. This effect of age is not clinically significant and does not warrant a dose adjustment.

Renal impairment

It was reported that, tadalafil exposure (AUC) approximately doubled in patients with mild (creatinine clearance 51 to 80ml/min) or moderate (creatinine clearance 31 to 50ml/min) renal impairment and in patients with end-stage renal disease on dialysis when they were administered a single-dose tadalafil (5-20 mg). In haemodialysis patients, C_{max} was reported to be 41% higher than that reported in healthy individuals. Haemodialysis contributes negligibly to tadalafil elimination.

Hepatic impairment

It was reported that tadalafil exposure (AUC) in patients with mild and moderate hepatic impairment (Child-Pugh Class A and B) is comparable to exposure in healthy individuals when a dose of 10mg is administered. There is limited information on the safety of tadalafil in patients with severe hepatic impairment (Child-Pugh Class C). There are no available information about the administration of once-a-day dosing of tadalafil to patients with hepatic impairment. If tadalafil is prescribed once-a-day, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician.

Patients with diabetes

Tadalafil exposure (AUC) in patients with diabetes was reported to be approximately 19% lower than the AUC value for healthy individuals. This difference in exposure does not warrant a dose adjustment.

INDICATIONS¹

Treatment of erectile dysfunction (ED) in adult males. In order for tadalafil to be effective for the treatment of ED, sexual stimulation is required.

Tadalafil is not indicated for use by women.

DOSE AND METHOD OF ADMINISTRATION ¹

Posology

Adult men

In general, the recommended dose is 10 mg taken prior to anticipated sexual activity and with or without food.

In those patients in whom tadalafil 10 mg does not produce an adequate effect, 20 mg might be tried. It may be taken at least 30 minutes prior to sexual activity.

The maximum dose frequency is once per day.

Tadalafil 10 and 20 mg is intended for use prior to anticipated sexual activity and it is not recommended for continuous daily use.

In patients who anticipate a frequent use of tadalafil (i.e., at least twice weekly), a once daily regimen with the lowest doses of tadalafil might be considered suitable, based on patient choice and the physician's judgement.

In these patients, the recommended dose is 5 mg taken once a day at approximately the same time of day. The dose may be decreased to 2.5 mg once a day based on individual tolerability.

The appropriateness of continued use of the daily regimen should be reassessed periodically.

Special Populations

Elderly Men

Dose adjustments are not required in elderly patients.

Men with renal impairment

Dose adjustments are not required in patients with mild to moderate renal impairment. For patients with severe renal impairment 10 mg is the maximum recommended dose. Once-a-day dosing of tadalafil is not recommended in patients with severe renal impairment.

Men with Hepatic Impairment

The recommended dose of Tadalafil is 10 mg taken prior to anticipated sexual activity and with or without food. There is limited clinical data on the safety of Tadalafil in patients with severe hepatic impairment (Child- Pugh Class C); if prescribed, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician. There are no available data about the administration of doses higher than 10 mg of tadalafil to patients with hepatic impairment. Once-a-day dosing has not been evaluated in patients with hepatic impairment; therefore if prescribed, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician.

Men with Diabetes

Dose adjustments are not required in diabetic patients.

Paediatric population

There is no relevant use of Tadalafil in the paediatric population with regard to the treatment of erectile dysfunction.

CONTRAINDICATIONS^{1, 2}

Hypersensitivity to the active substance or to any of the excipients.

Tadalafil was reported to augment the hypotensive effects of nitrates. This is thought to result from the combined effects of nitrates and tadalafil on the nitric oxide/cGMP pathway. Therefore, administration of tadalafil to patients who are using any form of organic nitrate is contraindicated (see **DRUG INTERACTIONS**).

Tadalafil should not be used in men with cardiac disease for whom sexual activity is inadvisable. Physicians should consider the potential cardiac risk of sexual activity in patients with pre-existing cardiovascular disease.

Tadalafil is contraindicated in the following groups of patients with cardiovascular disease:

- patients with myocardial infarction within the last 90 days
- patients with unstable angina or angina occurring during sexual intercourse
- patients with New York Heart Association Class 2 or greater heart failure in the last 6 months
- patients with uncontrolled arrhythmias, hypotension (<90/50 mmHg), or uncontrolled hypertension
- patients with a stroke within the last 6 months

Tadalafil is contraindicated in patients who have loss of vision in one eye because of non-arteritic anterior ischaemic optic neuropathy (NAION), regardless of whether this episode was in connection or not with previous PDE5 inhibitor exposure (see **WARNINGS AND PRECAUTIONS**).

The co-administration of PDE5 inhibitors, including tadalafil, with guanylate cyclase stimulators, such as riociguat, is contraindicated as it may potentially lead to symptomatic hypotension (see **DRUG INTERACTIONS**).

WARNINGS AND PRECAUTIONS ^{1, 2}

Before treatment with tadalafil

A medical history and physical examination should be undertaken to diagnose erectile dysfunction (ED) and determine potential underlying causes, before pharmacological treatment is considered.

Prior to initiating any treatment for ED, physicians should consider the cardiovascular status of their patients, since there is a degree of cardiac risk associated with sexual activity. Tadalafil has vasodilator properties, resulting in mild and transient decreases in blood pressure and as such potentiate the hypotensive effect of nitrates (see **CONTRAINDICATIONS**).

The evaluation of ED should include a determination of potential underlying causes and the identification of appropriate treatment following an appropriate medical assessment. It is not known if tadalafil is effective in patients who have undergone pelvic surgery or radical non-nerve-sparing prostatectomy.

Cardiovascular

Serious cardiovascular events, including myocardial infarction, sudden cardiac death, unstable angina pectoris, ventricular arrhythmia, stroke, transient ischemic attacks, chest pain, palpitations and tachycardia, have been reported in patients receiving tadalafil. Most of the patients in whom these events have been reported had pre-existing cardiovascular risk factors. However, it is not

possible to definitively determine whether these events are related directly to these risk factors, to tadalafil, to sexual activity or to a combination of these or other factors.

In patients receiving concomitant anti-hypertensive medicinal products, tadalafil may induce a blood pressure decrease. When initiating daily treatment with tadalafil, appropriate clinical considerations should be given to a possible dose adjustment of the anti-hypertensive therapy.

In patients who are taking alpha₁ blockers, such as doxazosin, concomitant administration of tadalafil may lead to symptomatic hypotension in some patients (see **DRUG INTERACTIONS**). Therefore, the combination of tadalafil and alpha blockers is not recommended.

Vision

Visual defects and cases of NAION have been reported in connection with the intake of tadalafil and other PDE5 inhibitors. The patient should be advised that in case of sudden visual defect, he should stop taking tadalafil and consult a physician immediately (see **CONTRAINDICATIONS**).

Renal and hepatic impairment

Due to increased tadalafil exposure (AUC), limited clinical experience and the lack of ability to influence clearance by dialysis, once-a-day dosing of tadalafil is not recommended in patients with severe renal impairment.

There are limited information on the safety of single-dose administration of tadalafil in patients with severe hepatic impairment (Child-Pugh Class C). Once-a-day administration has not been evaluated in patients with hepatic impairment. If tadalafil is prescribed, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician.

Priapism and anatomical deformation of the penis

Patients who experience erections lasting 4 hours or more should be instructed to seek immediate medical assistance. If priapism is not treated immediately, penile tissue damage and permanent loss of potency may result.

Tadalafil should be used with caution in patients with anatomical deformation of the penis (such as angulation, cavernosal fibrosis or Peyronie's disease), or in patients who have conditions which may predispose them to priapism (such as sickle cell anaemia, multiple myeloma or leukaemia).

Use with CYP3A4 inhibitors

Caution should be exercised when prescribing tadalafil to patients using potent CYP3A4 inhibitors (ritonavir, saquinavir, ketoconazole, itraconazole and erythromycin) as increased tadalafil exposure (AUC) has been reported if the medicinal products are combined (see **DOSE AND METHOD OF ADMINISTRATION** and **DRUG INTERACTIONS**).

Tadalafil and other treatments for erectile dysfunction

The safety and efficacy of combinations of tadalafil and other PDE5 inhibitors or other treatments for ED have not been studied. Therefore, the use of such combinations is not recommended.

Decreased or sudden hearing loss

Cases of sudden hearing loss have been reported after the use of tadalafil. Although other risk factors were present in some cases (such as age, diabetes, hypertension and previous hearing loss history) patients should be advised to stop taking tadalafil and seek prompt medical attention in the event of sudden decrease or loss of hearing.

Lactose

This product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicinal product.

DRUG INTERACTIONS ^{1, 2}

Interactions reported with tadalafil 10 mg and/or 20 mg are presented below.

Effects of other medicinal products on tadalafil

Increased Tadalafil AUC exposure:

Protease inhibitor i.e. Ritonavir, saquinavir and CYP3A4 inhibitors, i.e. erythromycin, clarithromycin, ketoconazole, itraconazole and grapefruit juice should be co-administered with caution as they would be expected to increase plasma concentrations of tadalafil. Consequently the incidence of the adverse reactions might be increased.

Reduced Tadalafil AUC exposure:

A CYP3A4 inducer, rifampicin, reduced tadalafil AUC exposure. This reduced exposure can be anticipated to decrease the efficacy of tadalafil; the magnitude of decreased efficacy is unknown. Other inducers of CYP3A4, such as phenobarbital, phenytoin, and carbamazepine, may also decrease plasma concentrations of tadalafil.

Effects of tadalafil on other medicinal products

Nitrates: augment the hypotensive effects of nitrates. Therefore, concomitant use of any form of organic nitrate is contraindicated.

Thus, in a patient prescribed any dose of tadalafil (2.5-20mg), where nitrate administration is deemed medically necessary in a life-threatening situation, at least 48 hours should have elapsed after the last dose of tadalafil before nitrate administration is considered. In such circumstances,

nitrates should only be administered under close medical supervision with appropriate haemodynamic monitoring.

Anti-hypertensives: augment blood pressure-lowering effect of doxazosin significantly when co-administered with tadalafil. This effect lasts at least twelve hours and may be symptomatic, including syncope. Therefore, this combination is not recommended (see **WARNINGS AND PRECAUTIONS**). Although these effects were not reported with alfuzosin or tamsulosin, caution should be exercised when using tadalafil in patients treated with any alpha-blockers, and notably in the elderly. Treatments should be initiated at minimal dosage and progressively adjusted.

Tadalafil had no clinically significant interaction with major classes of anti-hypertensive medicinal products including calcium channel blockers (amlodipine), ACE-inhibitors (enalapril), beta-adrenergic receptor blockers (metoprolol), thiazide diuretics (bendrofluazide), and angiotensin II receptor blockers (various types and doses, alone or in combination with thiazides, calcium channel blockers, beta blockers and/or alpha blockers).

In patients receiving concomitant anti-hypertensive medicinal products, tadalafil 20mg may induce a blood pressure decrease, which (with the exception of alpha blockers - see above) is, in general, minor and not likely to be clinically relevant.

No difference in adverse events was reported in patients taking tadalafil with or without anti-hypertensive medicinal products. However, appropriate clinical advice should be given to patients regarding a possible decrease in blood pressure when they are treated with anti-hypertensive medicinal products.

Riociguat: augment hypotensive effects of PDE5 inhibitor. No evidence of favourable clinical effect of the combination has been reported. Concomitant use of riociguat with PDE5 inhibitors, including tadalafil, is contraindicated (see **CONTRAINDICATIONS**).

Caution should be exercised when tadalafil is co-administered with:

- 5-alpha reductase inhibitors (5-ARIs): Finasteride
- CYP1A2 substrates (e.g. theophylline): minor increase in heart rate
- Ethinylestradiol and terbutaline

Co-administration with tadalafil did not affect the following substance:

Alcohol: did not augment blood pressure, but decrease in blood pressure produced by alcohol (vodka)

Aspirin: did not potentiate the increase in bleeding time caused by aspirin.

Anti-diabetic medicinal products: no specific interaction

Cytochrome P450 metabolised medicinal products: does not affect clearance of medicinal products metabolised by CYP450 isoforms including CYP3A4, CYP1A2, CYP2D6, CYP2E1, CYP2C9 and CYP2C19.

CYP2C9 substrates: did not affect changes in prothrombin time of warfarin (R-warfarin and S-warfarin).

SIDE EFFECTS/UNDESIRABLE EFFECTS ^{1, 2}

Summary of the safety profile

The most commonly reported adverse reactions in patients taking tadalafil for the treatment of ED or BPH were headache, dyspepsia, back pain and myalgia, in which the incidences increase with increasing dose of tadalafil. The adverse reactions reported were transient, and generally mild or moderate. The majority of headaches reported with tadalafil once-a-day dosing are experienced within the first 10 to 30 days of starting treatment.

Tabulated summary of adverse reactions

Very common, Common, Uncommon, Rare, Very Rare and Not known side effects presented below.

Organ System	Common	Uncommon	Rare
Immune system disorders		Hypersensitivity reactions	Angioedema ²
Nervous system disorders	Headache	Dizziness	Stroke ¹ (including haemorrhagic events), Syncope, Transient ischaemic attacks ¹ , Migraine ² , Seizures ² , Transient amnesia
Eye disorders		Blurred vision, Sensations described as eye pain	Visual field defect, Swelling of eyelids, Conjunctival hyperaemia, Non-arteritic anterior ischaemic optic neuropathy (NAION) ² , Retinal vascular occlusion ²
Ear and labyrinth disorders		Tinnitus	Sudden hearing loss
Cardiac disorders¹		Tachycardia, Palpitations	Myocardial infarction, Unstable angina pectoris ² , Ventricular arrhythmia ²
Vascular disorders	Flushing	Hypotension ³ , Hypertension	
Respiratory, thoracic and mediastinal disorders	Nasal congestion	Dyspnoea, Epistaxis	
Gastrointestinal disorders	Dyspepsia	Abdominal pain, Vomiting, Nausea, Gastro-oesophageal reflux	
Skin and subcutaneous		Rash	Urticaria, Stevens-Johnson

Organ System	Common	Uncommon	Rare
tissue disorders			syndrome ² , Exfoliative dermatitis ² , Hyperhydrosis (sweating)
Musculoskeletal, connective tissue and bone disorders	Back pain, Myalgia, Pain in extremity		
Renal and urinary disorders		Haematuria	
Reproductive system and breast disorders		Prolonged erections	Priapism, Penile haemorrhage, Haemospermia
General disorders and administration site conditions		Chest pain ¹ , Peripheral oedema, Fatigue	Facial oedema ² , Sudden cardiac death ^{1,2}

¹Most of the patients had pre-existing cardiovascular risk factors (see **WARNINGS AND PRECAUTIONS**).

²More commonly reported when tadalafil is given to patients who are already taking antihypertensive medicinal products.

Description of selected adverse reactions

ECG abnormalities, primarily sinus bradycardia, has been reported in patients treated with tadalafil once a day. Most of these ECG abnormalities were not reported to be associated with adverse reactions.

Other special populations

There is limited information in patients over 65 years of age receiving tadalafil either for the treatment of erectile dysfunction or the treatment of benign prostatic hyperplasia.

Diarrhoea was reported more frequently when tadalafil was administered for the treatment of erectile dysfunction in patients over 65 years of age. Dizziness and diarrhoea were reported more frequently when tadalafil 5 mg was administered for the treatment of benign prostatic hyperplasia in patients over 75 years of age.

USE IN SPECIAL POPULATIONS ^{1, 2}

Pregnancy

As a precautionary measure, it is preferable to avoid the use of Tadalafil during pregnancy.

Tadalafil is not indicated for use by women. There is no reported clinical experience on exposed pregnancies. No direct or indirect harmful effects with respect to pregnancy, embryonal/foetal development, parturition or postnatal development have been reported in animals.

Breastfeeding

Available pharmacodynamic/toxicological data in animals have shown excretion of tadalafil in milk. A risk to the suckling child cannot be excluded. Tadalafil should not be used during breast feeding.

Fertility

Effects were reported in dogs that might indicate impairment of fertility. Literature suggests that this effect is unlikely in humans, although a decrease in sperm concentration was reported in some men.

OVERDOSE ^{1, 2}

Similar adverse events were reported with tadalafil lower dose, single doses of up to 500 mg in healthy individuals and multiple daily doses of up to 100 mg in patients.

In cases of overdose, standard supportive measures should be adopted, as required.

Haemodialysis contributes negligibly to tadalafil elimination.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Tadalafil has negligible influence on the ability to drive or use machines. Dizziness has been reported with tadalafil, patients should be aware of how they react to tadalafil before driving or using machines.

STORAGE

Store below 30°C, protected from light.

SHELF LIFE

24 Months (2 Years)

AVAILABILITY

TALIS 20 (Tadalafil Tablets USP, 20 mg) available 04 tablets in a carton along with package insert. Each carton contains two blisters carrying two tablets each.

MANUFACTURED BY

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